

Yefei Zhang

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Education

University of Illinois Urbana–Champaign (UIUC) Champaign, IL | *Expected May 2027*
Ph.D. in Mathematics (Advisor: Daniel Cooney); M.S. in Mathematics, May 2023 — GPA: 3.83/4.0

Research: Nonlinear dynamics on networks, PDEs, stochastic processes, optimization, and machine learning

Colorado State University Fort Collins, CO | *May 2021*

B.S. in Mathematics and Chemistry — GPA: 4.0/4.0, *Summa Cum Laude*

Dual B.S. in Chemistry, East China Normal University (Shanghai, China); Exchange Student, Belmont University (2019)

Work Experience

AI Research Intern — TheWake Systems Shanghai, China | May–Aug 2026

- Built multimodal SFT and preference datasets for **RiverONE**, a **1.9B**-parameter scientific vision–language model (VLM) for quantum-calibration plot understanding — scientific reasoning, parameter extraction, and calibration diagnosis.
- Post-trained the model via supervised fine-tuning (SFT) and Mixed Preference Optimization (MPO); on the **QCalEval** benchmark, RiverONE retains $\geq 95\%$ of the performance of a closed-source domain baseline with $< 10\%$ of its parameters.
- Designed a variational-quantum-circuit (VQC) parameter generator producing layer-specific compensation for compressed ViT blocks.

Graduate Fellow — Institute for Pure & Applied Mathematics, UCLA Los Angeles, CA | Mar–Jun 2025

- Selected fellow for the semester-long program on non-commutative optimal transport; studied Wasserstein-type metrics for quantum states and operator algebras.
- Developed computational methods for entropy-regularized (Sinkhorn-type) transport — the optimization framework underlying modern generative modeling and distribution matching in ML.

Quantum Research Intern — ERROR CORP. College Park, MD | May–Aug 2024

- Applied reinforcement learning to discover compact, hardware-adapted encoders for the $[[5,1,3]]$ and $[[7,1,3]]$ codes, achieving 20–30% gate-count reduction.
- Developed and optimized fault-tolerant quantum circuits using **Quirk** and **PyZX**, applying ZX-calculus rewrite rules to preserve logical fidelity and error-patterns.

Teaching Assistant — Department of Mathematics, UIUC Urbana, IL | Jan 2022–Present

- Taught Calculus, Linear Algebra, and Differential Equations; designed **PrairieLearn** assignments for 100+ students.

Selected Projects & Research

Illinois Mathematics Lab — Numerical Modeling of FKPP on Networks Urbana, IL | Jan–May 2026

- Led an undergraduate team building **finite-difference** solvers for reaction–diffusion (Fisher–KPP) equations on metric graphs; estimated traveling- and spreading-wave speeds on star, bridge, and tadpole topologies.

Illinois Mathematics Lab — Symplectic Map Optimization Urbana, IL | Aug–Dec 2025

- Led a research team on symplectic map optimization, parameterizing map families and applying the Adam optimizer to discover high-efficiency embeddings; Benchmarked algorithmic performance across parameter regimes.

Publications & Preprints

- Yefei Zhang** — *Token Dynamics under Masked Attention: Graphon Continuum Limits, Spectral Thresholds, and Pattern Formation*. In preparation, 2026.
- Yuxuan Zhao, Kaisheng Zhu, **Yefei Zhang**, and Daniel B. Cooney — *Pattern Formation in a Spatial Public Goods Dilemma due to Diffusive or Directed Motion*. *Bulletin of Mathematical Biology*, **88**(10), 173, 2026. doi:10.1007/s11538-026-01733-0.
- Xindian Ma, Xinyu Long, **Yefei Zhang**, et al. — *RiverONE: Generating Knowledge-Intensive VLM by Simulated Quantum Machines*, arXiv:2606.29966, 2026. **Outstanding Paper Award**, WAICA Workshop on AI-Driven Quantum Computing Acceleration, World AI Conference, Shanghai (Jul 2026).
- Amy Bradu, David (Rui) Sang, Stanley Combes, **Yefei Zhang**, Daniel B. Cooney — *Success-Driven Migration on Graphs and Graphons: Continuum Limits and Pattern Formation*. Under review, 2026.
- Yefei Zhang**, Daechol Kim, Yuxuan Zhao, Daniel B. Cooney — *The Keller–Segel Model for Chemotaxis on Metapopulation Graphs and Graphons*. Under review, 2025.

Skills

Programming & ML: Python (PyTorch, NumPy, SciPy, Pandas, scikit-learn, Matplotlib), C++, MATLAB, Git, Linux

Quantitative & Mathematical: Machine Learning, Optimization, Numerical Analysis, Stochastic Processes, PDEs, Quantum Computing, Dynamical Systems

Languages: English (Fluent), Chinese (Native)